Pamela Ballesteros

MBAN 5540 -Prescriptive Analytics & Optimization
Assignment II

1. What is the production plan and total profit? Assume that fractional output is acceptable.
• C1: # of small canoes = 13.4066
• C2: # of regular canoes = 65.8022
• K1: # of one-person kayaks = 0
• K2: # of two-person kayaks = 57.1429
• D: #of dinghies = 10

Objective function:
Max 32C1 + 41C2 + 42K1 + 55K2 + 25D
=32 (13.4066) + 41 (65.8022) + 42 (0) + 55 (57.1429) + 25 (10) = 6519.76
Total Profit = 6519.76

2. Would the solution change if the minimum number of canoes made was increased to 60? If the minimum number of kayaks was increased to 60? If the minimum number of dinghies was increased to 20?

a. Minimum canoes increase to 60

Constraint of Min Canoe Total:

- RHS = 50, Final Value = 79.2087912 (not binding)
- Allowable Increase = 29.2087912
- RHS can increase up to 50 + 29.2087912 = 79.2087912 with no basis change

60 ≤ 79.2087912, the change is within the allowable increase, so solution and profit do not change with shadow price being 0 as well.

b. Minimum kayaks increase to 60

Constraint of Min Kayak Total:

- RHS = 50, Final Value = 57.1428571 (7.1428571)
- Allowable Increase = 7.1428571
- Max RHS without changing basis: 50 + 7.14285714 = 57.14285714

60 ≥ 57.1429, this is beyond allowable increase, which means the solution will change and profit will decrease.

c. Minimum dinghies increase to 20
   - Constraint Min Dinghy Total is binding at 10 (Final Value = RHS =10)
   - Shadow price = -3.12307692
   - Allowable Increase = 49.962406, increasing the RHS by 10 is within range**. However, because an increase of production of dinghies would result to a decrease by $3.12307692 to the profit, the solution will change.**

Z = (-3.12307692) (20-10) = -31.2307692

New profit: Z=6519.7582391 – 31.2307692 = 6488.53

3. It has been determined that the oven used to heat the kayak mold must be shut down in one month, leaving only 150 hours of time. Will this affect the solution or the profit?
Constraint of Kayak oven (≤ 240)

- Final Value = 131.428571
- RHS = 240
- Allowable Decrease =108.571429

Reducing RHS to 150 is a decrease of 90 (240-150), which is within allowable decrease. Also, even after the cut, it remains non-binding (131.42 ≤ 150). No change to solution or profit.

4. Extra fitting time can be obtained by hiring someone at a minimum wage of $10.85. Should we hire someone? If so, for how many hours?
Fittings total constraints (binding)

- Shadow price = 15.6923077 (profit gained per/hr)
- Allowable increase = 3.48571429 hours
- Cost per hour = 10.85

Fitting time is binding, with shadow price of $15.6923/hour. Since this exceeds the hiring cost of $10.85/hour, purchasing more fitting time increases net profit. The shadow price is valid for up to 3.4857 additional hours, giving an estimated net gain of (15.6923 −10.85) × 3.4857 = $16.88.
In conclusion, we should hire for the maximum supported hour from the report which is 3.4857 hours.

5. Suppose they can also get someone to help finish at $12 per hour. Should we hire someone? If so, for how many hours?

Constraint of Finishing Total is not binding and has a shadow price of 0, meaning an extra hour of finishing time would not increase profit at the current optimal solution. Since it would still cost $12 per hour, hiring additional finishing time would reduce profit.
Therefore, it is not advisable to hire for finishing time.

6. Management has noted that the plan calls for producing all products except the single kayak. What would be the impact on profits if they decided to make a single kayak?

- Single Kayak Final Value = 0
- Reduced Cost = -4.55238095

Interpretation for a maximation problem:
Profit impact for making 1 unit of Single Kayak (k1)
= 6519.7582 -4.5524 = 6515.2059

Since this is a maximation problem, forcing our model to produce K1, with a negative reduced cost of -4.5524, this would decrease the total profit.

7. Dinghies are only being made because they must deliver 10 of them. What would the profit per dinghy need to be to make us build them, given that they increase total profits.

- Constraint Min Dinghy Total is binding at 10 (Final Value = RHS =10)
- Shadow price = -3.12307692
- Objective Coefficient = 25

The dinghy profit must at least cover the opportunity cost before we can produce it voluntarily instead of being forced (D ≥ 10 units).

=25 + 3.12307692 = 28.12307692 or **$28.12 per dinghy**

8. RBB has not made racing kayaks for many years but still has the mold to produce them. To make a racing kayak, they would need 2.4 hours of mold prep time and 2.1 hours of mold oven time. They would also need 2.1 hours of cutting/finishing time and 2.5 hours for installation of fittings. How much profit would a racing kayak need to generate to make it profitable enough to consider?

By evaluating the production of racing kayaks again, we use the shadow prices (dual values) of the resources it consumes. Only binding resources have non-zero shadow prices.

Racing Kayak will use:

- Mold prep = 2.4 hrs Shadow Price = 6.01465201
- Oven 2.1 hrs Shadow Price = 0
- Finishing 2.1 hrs Shadow Price = 0
- Fittings 2.5 hrs Shadow Price = 15.6923077

Minimum profit threshold:

= 2.4 (6.01465201) + 2.1 (0) + 2.5 (15.6923077) = 53.67

Therefore, the racing kayak must generate at least $53.67 profit per unit to be worth considering.